



U.S. Department
of Transportation

**Pipeline and Hazardous
Materials Safety
Administration**

March 06, 2025

1200 New Jersey Avenue, SE
Washington, DC 20590

DOT-SP 7737
(SEVENTEENTH REVISION)

EXPIRATION DATE: 2029-02-28

(FOR RENEWAL, SEE 49 CFR 107.109)

1. GRANTEE: Catalina Cylinders, Inc.
Garden Grove, CA
2. PURPOSE AND LIMITATIONS:
 - a. This special permit authorizes the manufacture, mark, sale, and use of a non-DOT specification cylinder conforming to all regulations applicable to a DOT specification 3E cylinder, except as specified herein, for the transportation in commerce of the materials authorized by this special permit. This special permit provides no relief from the Hazardous Materials Regulations (HMR) other than as specifically stated herein. The most recent revision supersedes all previous revisions.
 - b. The safety analyses performed in development of this special permit only considered the hazards and risks associated with transportation in commerce.
 - c. In accordance with 49 CFR 107.107(a) party status may not be granted to a manufacturing permit. These packagings may be used in accordance with 49 CFR 173.22a.
3. REGULATORY SYSTEM AFFECTED: 49 CFR Parts 106, 107 and 171-180.
4. REGULATIONS FROM WHICH EXEMPTED: 49 CFR §§ 173.201(c), 173.192, 173.302a(a), 173.304a(a) and (4), and 173.337 in that a non-DOT specification cylinder is not authorized, except as specified herein.

5. BASIS: This special permit is based on the application of Catalina Cylinders, Inc. dated January 28, 2025, submitted in accordance with § 107.109.
6. HAZARDOUS MATERIALS (49 CFR § 172.101):

Hazardous Material Description			
Proper Shipping Name	Hazard Class/ Division	Identification Number	Packing Group
Any hazardous material for which DOT specification 3AL cylinder is prescribed or authorized in 49 CFR Part 173	2.1, 2.2, 2.3 and 3	As appropriate	As appropriate

7. SAFETY CONTROL MEASURES:

- a. PACKAGING: Prescribed packaging is a non-DOT specification seamless aluminum cylinder constructed in accordance with the design calculations dated: November 30, 1998, Part No. 9001; February 5, 1999, Part No. 9002; December 4, 1998, Part No. 9003; October 8, 1998, Part No. 9070; March 24, 1998 Part No. 9071; and October 8, 1998, Part No. 9072, on file with the Office of Hazardous Materials Safety (OHMS) and conforming with the specifications for DOT 3E cylinder (§§ 178.35 and 178.42) except as follows:

§ 178.35(b) *Inspections and analysis*.

Chemical analyses and tests as specified must be made within the United States. Inspections and verifications must be performed by an independent inspection agency approved in writing by the Associate Administrator, in accordance with § 107.803(b).

§ 178.35(f) *Markings*.

- (1) The cylinder must be marked with the following information: (see paragraph 8.i. of this special permit for a marking exception)

DOT-SP 7737-1800
M1039
MO-YR
ALUMINUM ALLOY

- (2) Where: MO-YR is the month and year of test.

§ 178.35(g) *Inspector's report.*

Applies except that the report must be modified as appropriate to accommodate 6061-T6 aluminum cylinder. The inspectors report on the first production lot must be submitted to the OHMS prior to initial shipment.

§ 178.42(a) *Type, size and service pressure.*

The packaging must be a seamless aluminum cylinder.
A spun cylinder is not authorized.

§ 178.42(b) *Material.*

Aluminum used must be alloy 6061 and T-6 temper designated by the Aluminum Association with the following properties:

CHEMICAL COMPOSITION LIMITS⁽¹⁾

Aluminum Alloy Designat-ion									Other ⁽²⁾		
	Si	Fe	Cu	Mn	Mg	Cr	Zn	Ti	Each	Total	A1
6061	0.40-0.80	0.70 Max.	0.15-0.40	0.15 Max.	0.8-1.2	0.04-0.35	0.25 Max.	0.15 Max.	0.05 Max.	0.15 Max.	Remain-der

MECHANICAL PROPERTY LIMITS

Alloy and Temper	Tensile Strength - psi		Elongation Percent Minimum in 2" or 4D
	Ultimate-Minimum	Yield-Minimum	
6061-T6	38,000	35,000	10

(1) ASTM B 221-72 Standard Specification for Aluminum Alloy extruded Bars Rods, Shapes, and Tubes, Table 1 Chemical Composition Limits.

(2) Analysis is regularly made only for the elements for which specific limits are shown, except for unalloyed aluminum. If, however, the presence of other elements is suspected to be, or in the course of routine analysis is indicated to be in excess of specified limits, further analysis is made to determine that these other elements are not in excess of the amount specified.

§ 178.42(c) *Identification of material.*

Material must be identified by any suitable method.

§ 178.42(d) *Manufacture.*

* * *

(Add) Cylinder shells must be manufactured by the Impact Extrusion Method and completed cylinder must be heat treated to the T-6 temper. Stress at minimum prescribed test pressure must not exceed 21,000 psi when calculated using the formula specified in § 178.36(f).

§ 178.42(e) *Openings in cylinders and connections.*

(1) * * *

(2) * * *

(3) Straight threads having at least 6 engaged threads are authorized. Straight threads must have a tight fit and a calculated shear strength at least 10 times the test pressure of the cylinder. Gaskets, adequate to prevent leakage, are required.

b. TESTING:

(1) *Design qualification tests.*

Prior to the initial shipment of any specific cylinder design, qualification tests as prescribed in this paragraph must have been performed and results submitted to the OHMS. All cylinders used for design qualification test must be fabricated on the same equipment and subjected to the same processes as is used to produce cylinders intended for charging and shipment. All tests must be witnessed by an independent inspector.

(i) Cycling test must be performed by hydrostatically pressuring the cylinder between approximately zero psig and the designated pressure at a rate not in excess of 10 cycles per minute. All cylinders used in cycle tests must be destroyed.

(ii) Three representative cylinders must be cycle tested at ambient temperature, without showing evidence of distortion, deterioration or failure, as follows:

First pressurize from 0 to service pressure for 100,000 cycles. After successfully passing this test, the cylinder must be pressurized to burst and the burst pressure recorded.

(2) *Physical tests.*

(i) The yield strength, tensile strength and elongation must be determined as prescribed in § 178.37(k).

(ii) Ultimate tensile strength, yield strength and elongation must confirm to at least the values found in § 178.42(b) of this special permit.

(3) *Flattening test.*

(i) One cylinder out of each lot of 200 cylinders or less must be flattened between wedge shaped fixtures, 60° included angle, edges rounded as follows:

Cylinder wall thickness	Radius
under - .150	.500
.150 - .249	.875
.250 - .349	1.500

(ii) Cylinder must be capable of being flattened to 9 times the wall thickness without cracking. Flattening to continue until cracking occurs and the point recorded on the report. Any cracking occurs must be longitudinal to the longitudinal axis of the cylinder.

(4) *Burst test.*

One cylinder taken at random out of each lot of cylinders must be hydrostatically tested to destruction. Burst pressure must be at least 3.33 times marked service pressure.

8. SPECIAL PROVISIONS:

a. In accordance with the provisions of Paragraph (b) of § 173.22a, persons may use the packaging authorized by this special permit for the transportation of the hazardous materials specified in paragraph 6, only in conformance with the terms of this special permit.

- b. A person who is not a holder of this special permit, but receives a package covered by this special permit, may reoffer it for transportation provided no modifications or changes are made to the package and it is offered for transportation in conformance with this special permit and the HMR.
 - c. A current copy of this special permit must be maintained at each facility where the package is offered or reoffered for transportation.
 - d. A current copy of this special permit must be maintained at each facility where the package is manufactured under this special permit. It must be made available to a DOT representative upon request.
 - e. The cylinder must be shipped in strong outer packagings in accordance with § 173.301(a)(9).
 - f. Except as provided for in this special permit, a cylinder that is to be used as a tire inflator, and only when installed in the trunk of a motor vehicle, is excepted from 49 CFR Parts 171 - 180.
 - g. Flammable (Division 2.1) and toxic (Division 2.3) gases are not authorized for transportation by cargo vessel or by aircraft.
 - h. Transportation of oxygen by aircraft is only authorized when in accordance with § 175.501.
 - (i) Cylinders marked with the symbol CLIFFDIV prior to June 9, 2017, may continue in use under the terms of this special permit.
9. MODES OF TRANSPORTATION AUTHORIZED: Motor vehicle, rail freight, cargo vessel and cargo aircraft only (see paragraphs 8.g and 8.h for restrictions).
10. MODAL REQUIREMENTS: A current copy of this special permit must be carried aboard each cargo vessel, aircraft or motor vehicle used to transport packages covered by this special permit. The shipper must furnish a current copy of this special permit to the air carrier before or at the time the shipment is tendered.
11. COMPLIANCE: Failure by a person to comply with any of the following may result in suspension or revocation of this special permit and penalties prescribed by the Federal hazardous materials transportation law, 49 U.S.C. 5101 et seq:
- o All terms and conditions prescribed in this special permit and the Hazardous Materials Regulations, Parts 171-180.

- o Persons operating under the terms of this special permit must comply with the security plan requirement in Subpart I of Part 172 of the HMR, when applicable.
- o Registration required by § 107.601 et seq., when applicable.

Each “Hazmat employee”, as defined in § 171.8, who performs a function subject to this special permit must receive training on the requirements and conditions of this special permit in addition to the training required by §§ 172.700 through 172.704.

No person may use or apply this special permit, including display of its number, when the special permit has expired or is otherwise no longer in effect.

Under Title VII of the Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (SAFETEA-LU) “The Hazardous Materials Safety and Security Reauthorization Act of 2005” (Pub. L. 109-59), 119 Stat. 1144 (August 10, 2005), amended the Federal hazardous materials transportation law by changing the term “exemption” to “special permit” and authorizes a special permit to be granted up to two years for new special permits and up to four years for renewals.

12. **REPORTING REQUIREMENTS:** Shipments or operations conducted under this special permit are subject to the Hazardous Materials Incident Reporting requirements specified in 49 CFR §§ 171.15 - Immediate notice of certain hazardous materials incidents, and 171.16 - Detailed hazardous materials incident reports. In addition, the grantee(s) of this special permit must notify the Associate Administrator for Hazardous Materials Safety, in writing, of any incident involving a package, shipment or operation conducted under terms of this special permit.

Issued in Washington, D.C.:



for William Schoonover
Associate Administrator for Hazardous Materials Safety

Address all inquiries to: Associate Administrator for Hazardous Materials Safety, Pipeline and Hazardous Materials Safety Administration, U.S. Department of Transportation, East Building PHH-13, 1200 New Jersey Avenue, Southeast, Washington, D.C. 20590.

Copies of this special permit may be obtained by accessing the Hazardous Materials Safety Homepage at <https://www.phmsa.dot.gov/approvals-and-permits/hazmat/special-permits-search>. Photo reproductions and legible reductions of this special permit are permitted. Any alteration of this special permit is prohibited.

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